T.I.P. TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES

COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/11/2026	**DATE SUBMITTED:**	07/11/2026
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ACTIVITY NO. 1.1 BASIC C++ PROGRAMMING

1. PROCEDURE OUTPUT

What procedures did you perform?

5. Procedure

ILO A: Create Code That Follows the Basic C++ Code Structure

For this activity, you have to demonstrate the use of a **function prototype**. The section on class declaration and method definition will be used for the function prototype and the function will be defined in the follow method definition section after the main function.

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Note:
- The driver program must call each function.
- The definitions must be after the main function.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

In this section, the initial implementation for a class **triangle** will be implemented. The step-by-step procedure is shown below:

Step 1. Include the necessary header files. For this one, we only need `#include <iostream>`

Step 2. Create the triangle class. Assign it with private variables: totalAngle, angleA, angleB, and angleC.

```
class Triangle{
private:
    double totalAngle, angleA, angleB, angleC;
```

Step 3. We then create public methods. The constructor must allow for creation of the object with 3 initial angles to be stored in our previously defined variables `angleA, angleB` and `angleC`. Another method has to be made if the user wants to change the initial values, this will also accept 3 arguments to change the values in `angleA, angleB` and `angleC`. Lastly, a function to validate whether the given values make our shape an actual triangle.

```
public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
```

Step 4. Define the methods.

```
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
```

Step 5. Create the driver code.

```
int main(){
    //driver code
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

Include the output of running this code in section 6. Note your observations and comments.

6. Output

Sections	Answer
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	
Method Definition	

Table 1-1. C++ Structure Code for Answer

Table 1-2. ILO B output observations and comments.

7. Supplementary Activity

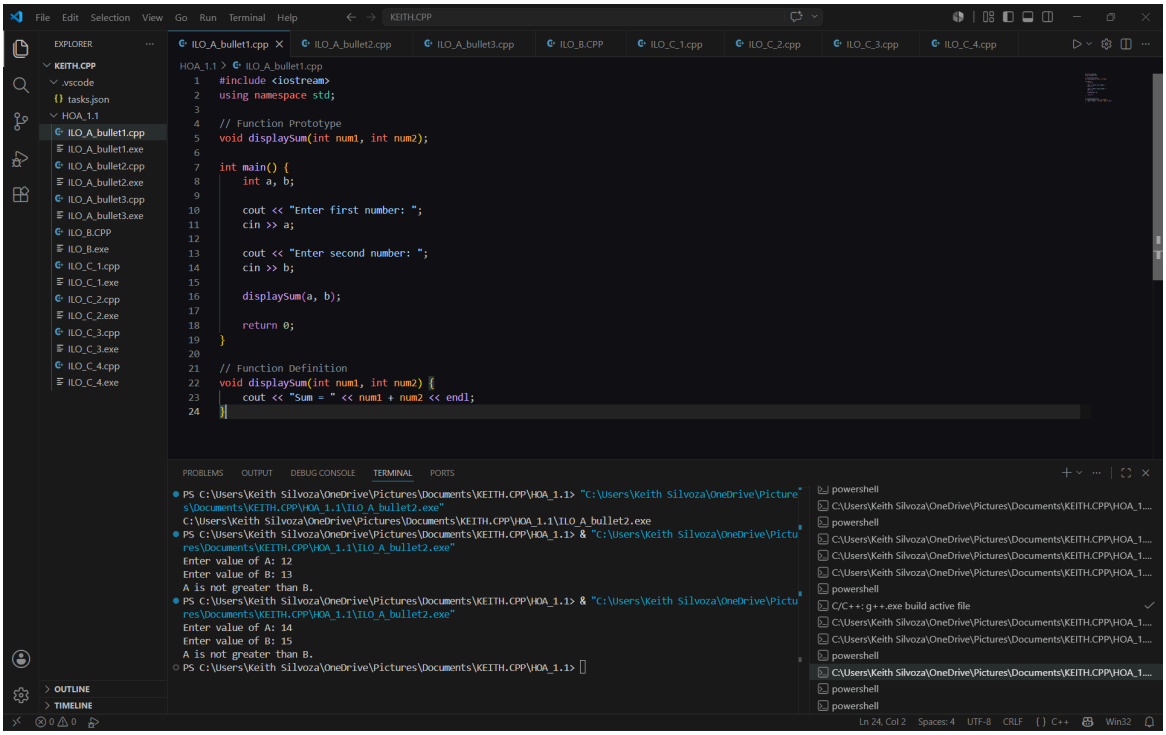
ILO C: Solve Different Problems using the C++ Programming Language

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.
2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
3. Create a C++ program that has a function that will calculate the distance between two points.
4. Modify the code given in ILO B and add the following functions:
    a. A function to compute for the area of a triangle
    b. A function to compute for the perimeter of a triangle
    c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO-A



```
#include <iostream>
using namespace std;

// Function Prototype
void displaySum(int num1, int num2);

int main() {
    int a, b;

    cout << "Enter first number: ";
    cin >> a;

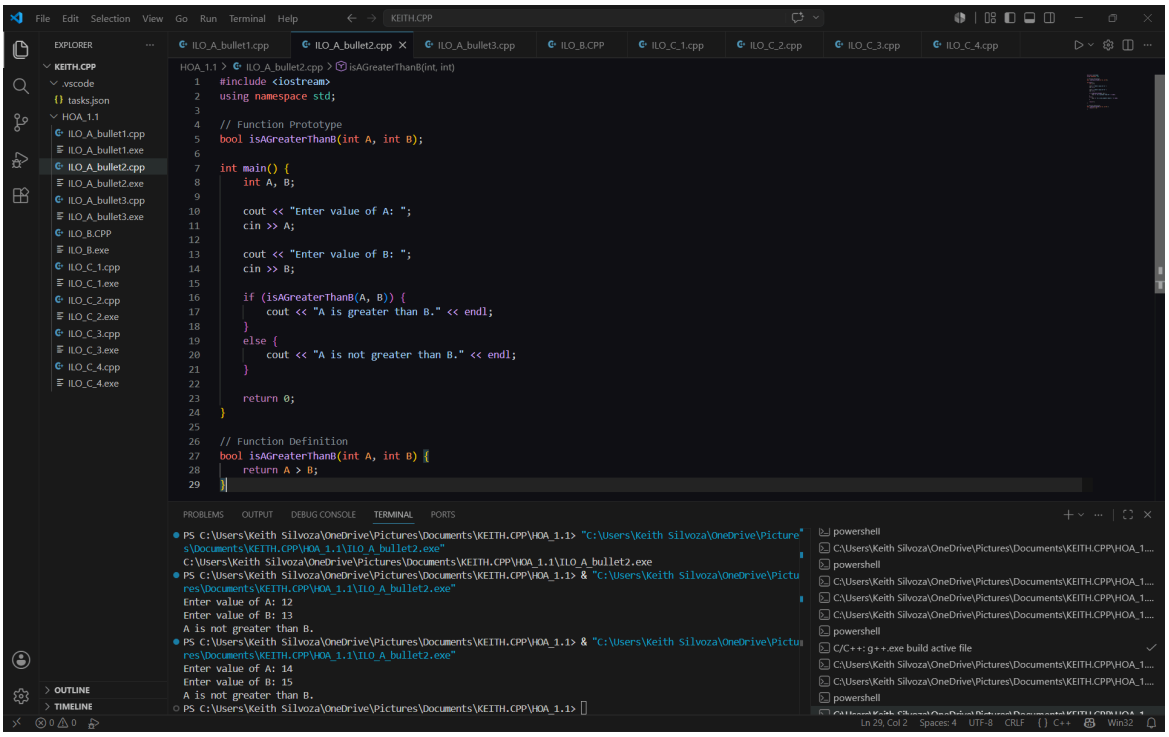
    cout << "Enter second number: ";
    cin >> b;

    displaySum(a, b);

    return 0;
}

// Function Definition
void displaySum(int num1, int num2) {
    cout << "Sum = " << num1 + num2 << endl;
}
```

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe"
C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe"
Enter value of A: 12
Enter value of B: 13
A is not greater than B.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe"
Enter value of A: 14
Enter value of B: 15
A is not greater than B.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```



```
#include <iostream>
using namespace std;

// Function Prototype
bool isAGreaterThanB(int A, int B);

int main() {
    int A, B;

    cout << "Enter value of A: ";
    cin >> A;

    cout << "Enter value of B: ";
    cin >> B;

    if (isAGreaterThanB(A, B)) {
        cout << "A is greater than B." << endl;
    }
    else {
        cout << "A is not greater than B." << endl;
    }

    return 0;
}

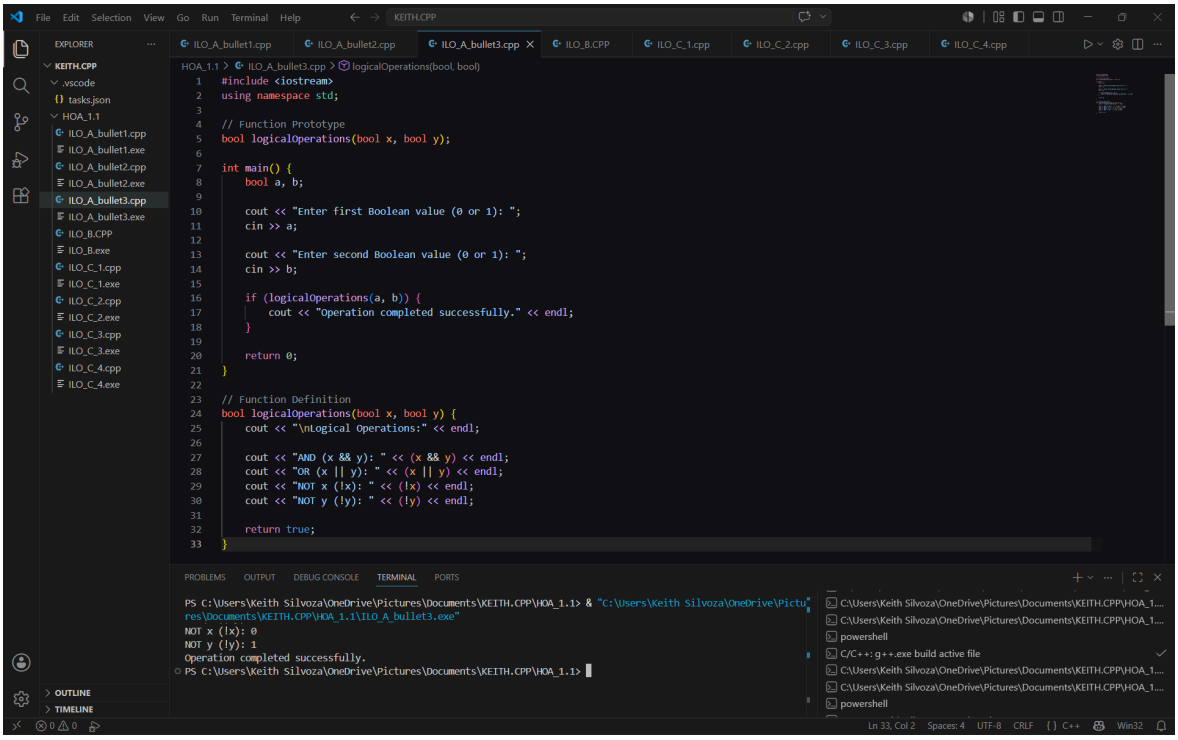
// Function Definition
bool isAGreaterThanB(int A, int B) {
    return A > B;
}
```

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe"
C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe"
Enter value of A: 12
Enter value of B: 13
A is not greater than B.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet2.exe"
Enter value of A: 14
Enter value of B: 15
A is not greater than B.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```

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```cpp
#include <iostream>
using namespace std;

// Function Prototype
bool logicalOperations(bool x, bool y);

int main() {
    bool a, b;

    cout << "Enter first Boolean value (0 or 1): ";
    cin >> a;

    cout << "Enter second Boolean value (0 or 1): ";
    cin >> b;

    if (logicalOperations(a, b)) {
        cout << "Operation completed successfully." << endl;
    }

    return 0;
}

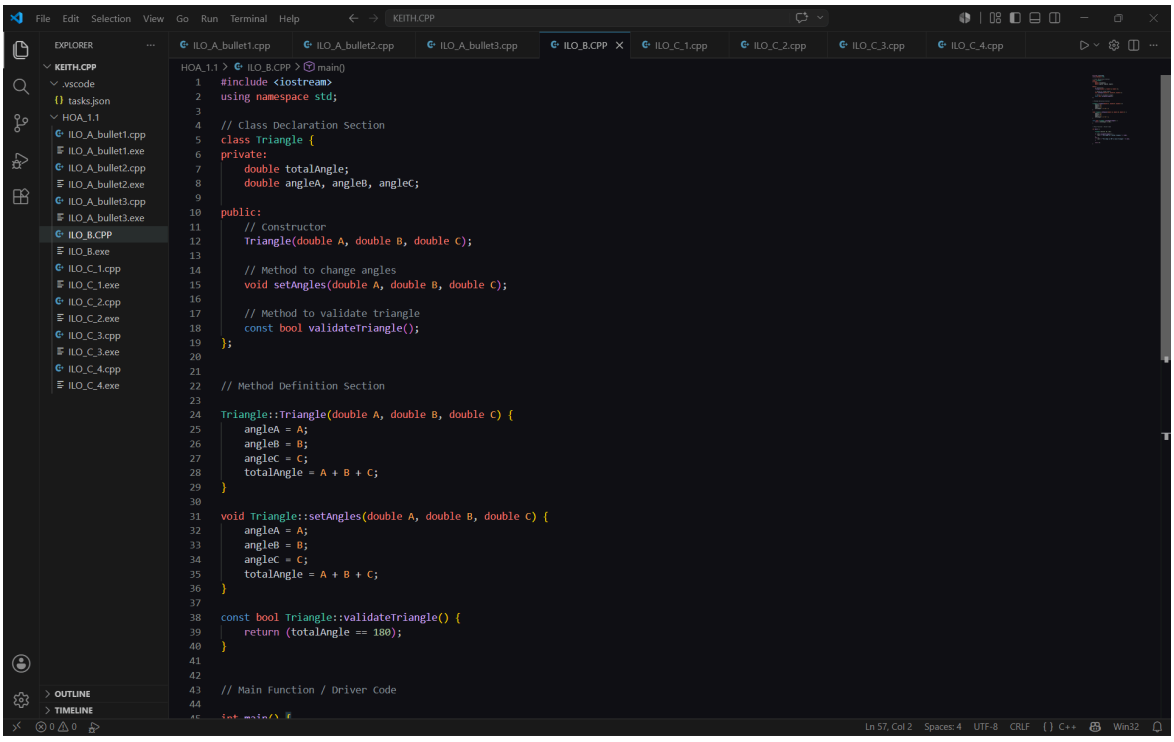
// Function Definition
bool logicalOperations(bool x, bool y) {
    cout << "\nLogical Operations:" << endl;

    cout << "AND (x && y): " << (x && y) << endl;
    cout << "OR (x || y): " << (x || y) << endl;
    cout << "NOT x (!x): " << (!x) << endl;
    cout << "NOT y (!y): " << (!y) << endl;

    return true;
}
```

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_A_bullet3.exe"
NOT x (!x): 0
NOT y (!y): 1
Operation completed successfully.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```

ILO-B



```cpp
#include <iostream>
using namespace std;

// Class Declaration Section
class Triangle {
private:
    double totalAngle;
    double angleA, angleB, angleC;

public:
    // Constructor
    Triangle(double A, double B, double C);

    // Method to change angles
    void setAngles(double A, double B, double C);

    // Method to validate triangle
    const bool validateTriangle();
};


// Method Definition Section

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

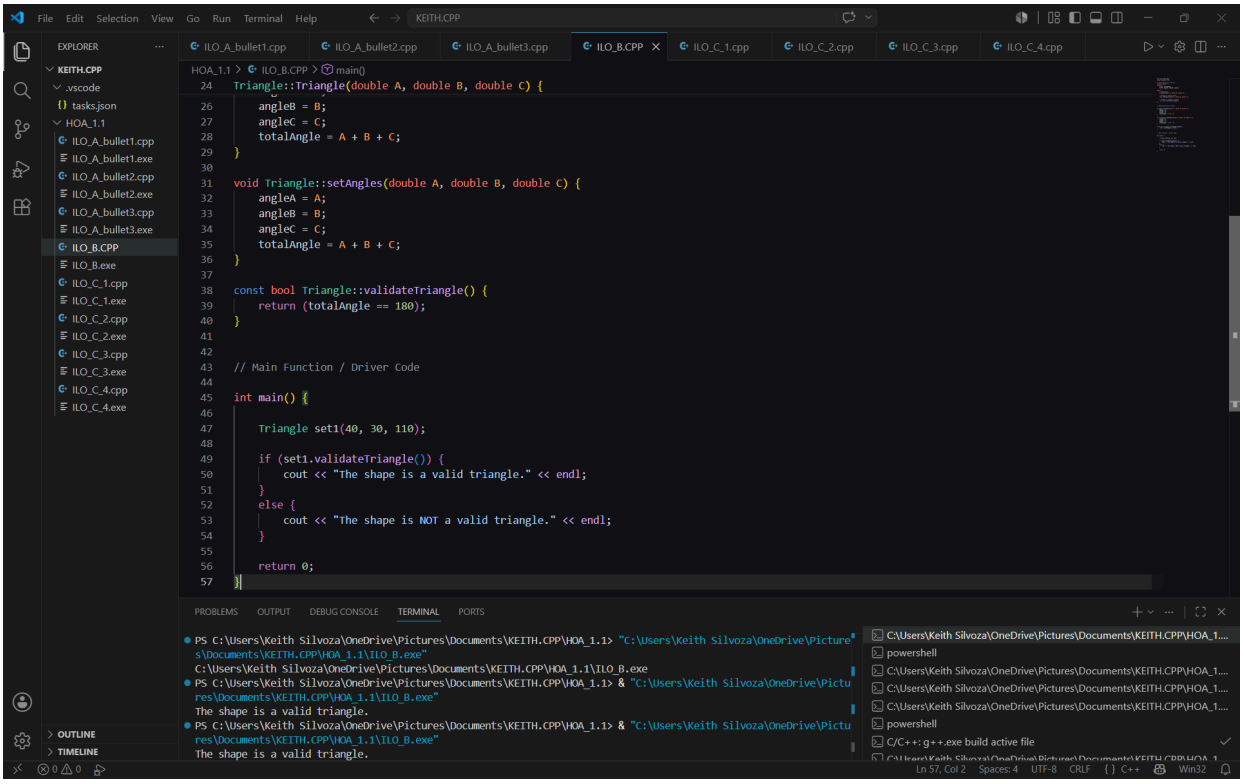
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}


// Main Function / Driver Code
```

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```cpp
Triangle::Triangle(double A, double B, double C) {
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}


// Main Function / Driver Code

int main() {

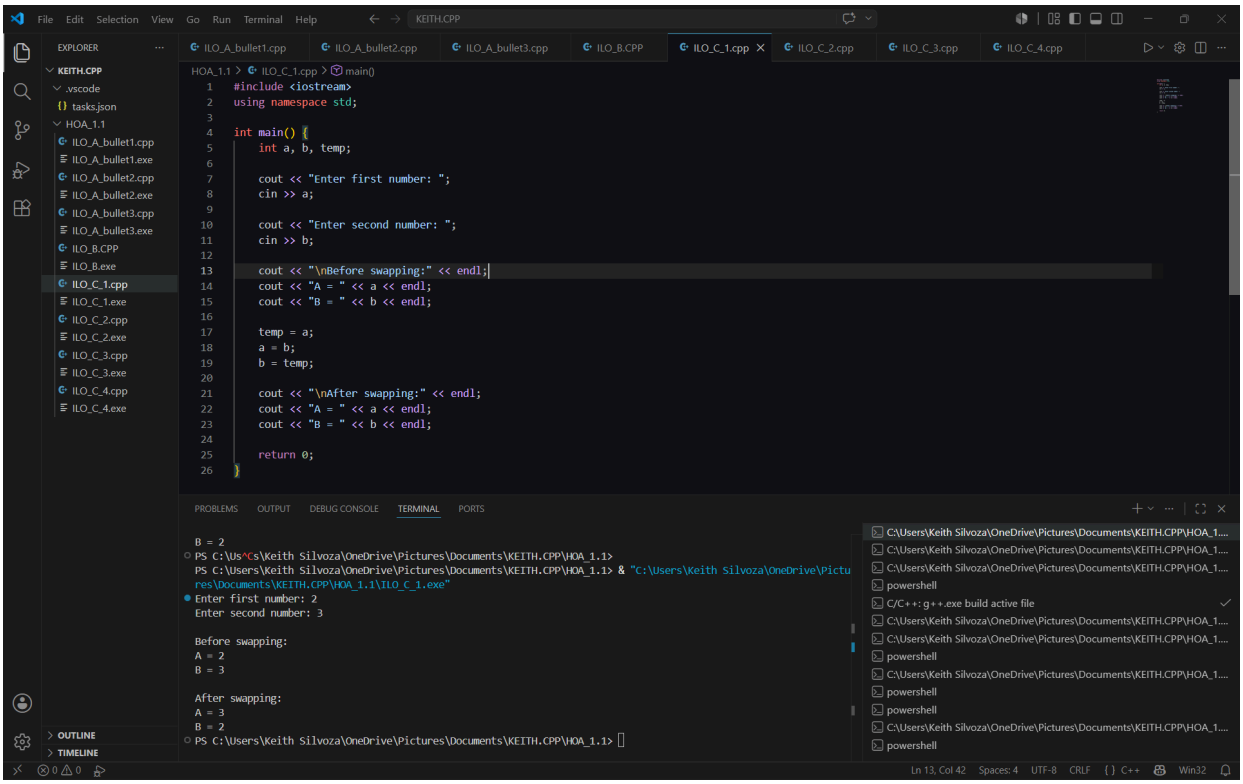
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle." << endl;
    }
    else {
        cout << "The shape is NOT a valid triangle." << endl;
    }

    return 0;
}
```

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_B.exe"
C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_B.exe
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_B.exe"
The shape is a valid triangle.
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_B.exe"
The shape is a valid triangle.
```

ILO-C



```c
#include <iostream>
using namespace std;

int main() {
    int a, b, temp;

    cout << "Enter first number: ";
    cin >> a;

    cout << "Enter second number: ";
    cin >> b;

    cout << "\nBefore swapping:" << endl;
    cout << "A = " << a << endl;
    cout << "B = " << b << endl;

    temp = a;
    a = b;
    b = temp;

    cout << "\nAfter swapping:" << endl;
    cout << "A = " << a << endl;
    cout << "B = " << b << endl;

    return 0;
}
```

```
B = 2
PS C:\Us^Cs\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_C_1.exe"
Enter first number: 2
Enter second number: 3

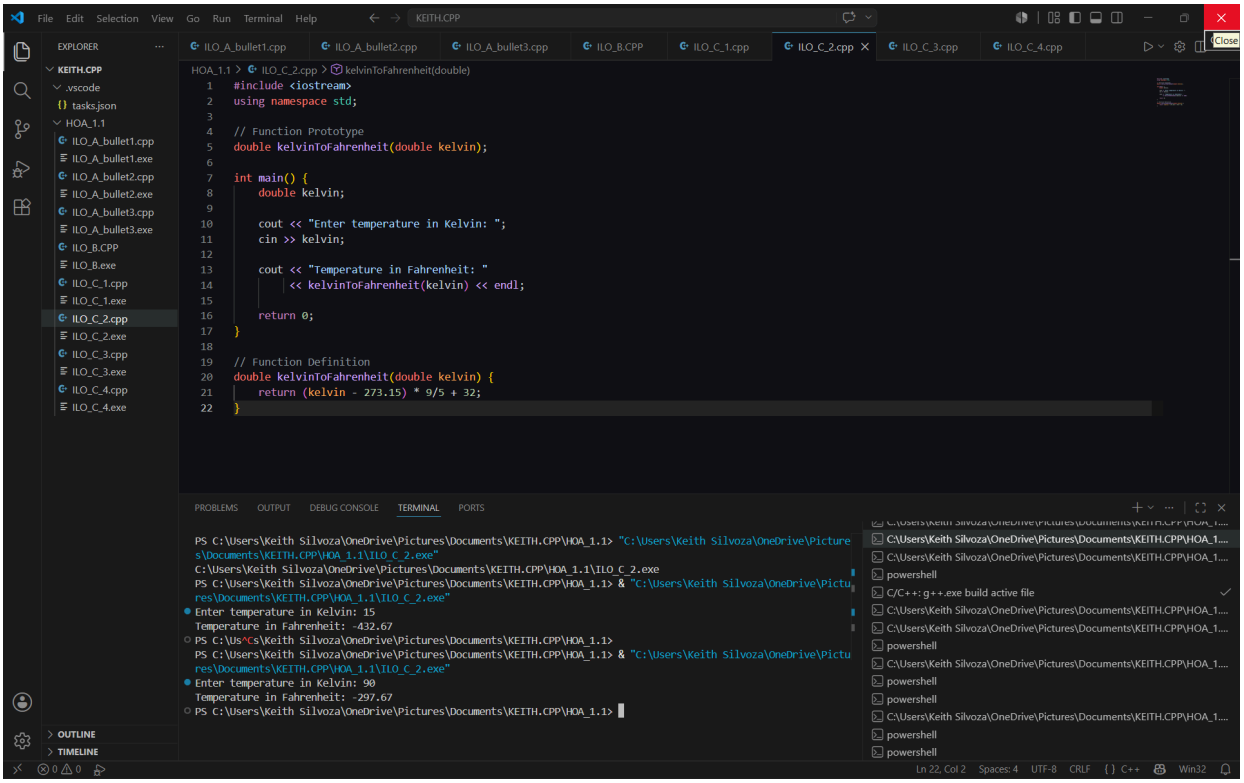
Before swapping:
A = 2
B = 3

After swapping:
A = 3
B = 2
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```

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```cpp
#include <iostream>
using namespace std;

// Function Prototype
double kelvinToFahrenheit(double kelvin);

int main() {
    double kelvin;

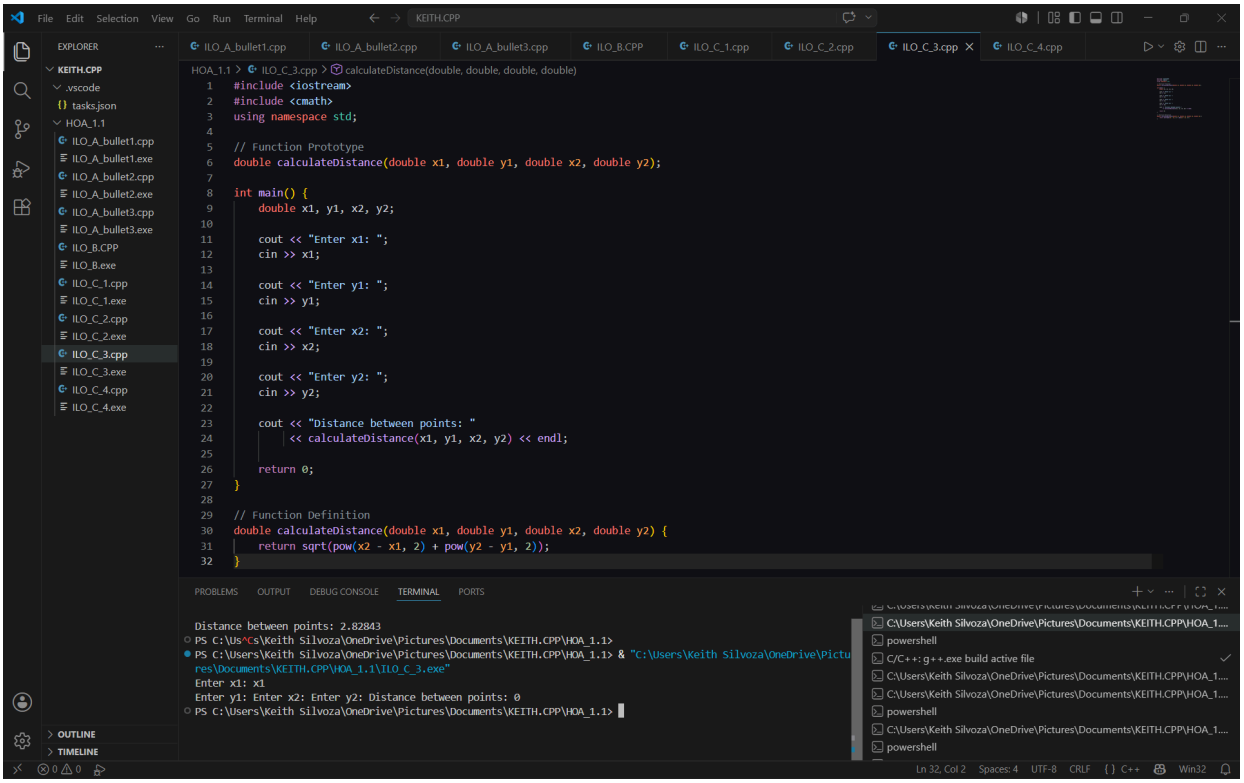
    cout << "Enter temperature in Kelvin: ";
    cin >> kelvin;

    cout << "Temperature in Fahrenheit: "
         << kelvinToFahrenheit(kelvin) << endl;

    return 0;
}

// Function Definition
double kelvinToFahrenheit(double kelvin) {
    return (kelvin - 273.15) * 9/5 + 32;
}
```

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_C_2.exe"
C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_C_2.exe
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_C_2.exe"
Enter temperature in Kelvin: 15
Temperature in Fahrenheit: -432.67
PS C:\Us^Cs\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "c:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_C_2.exe"
Enter temperature in Kelvin: 90
Temperature in Fahrenheit: -297.67
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```



```cpp
#include <iostream>
#include <cmath>
using namespace std;

// Function Prototype
double calculateDistance(double x1, double y1, double x2, double y2);

int main() {
    double x1, y1, x2, y2;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    cout << "Distance between points: "
         << calculateDistance(x1, y1, x2, y2) << endl;

    return 0;
}

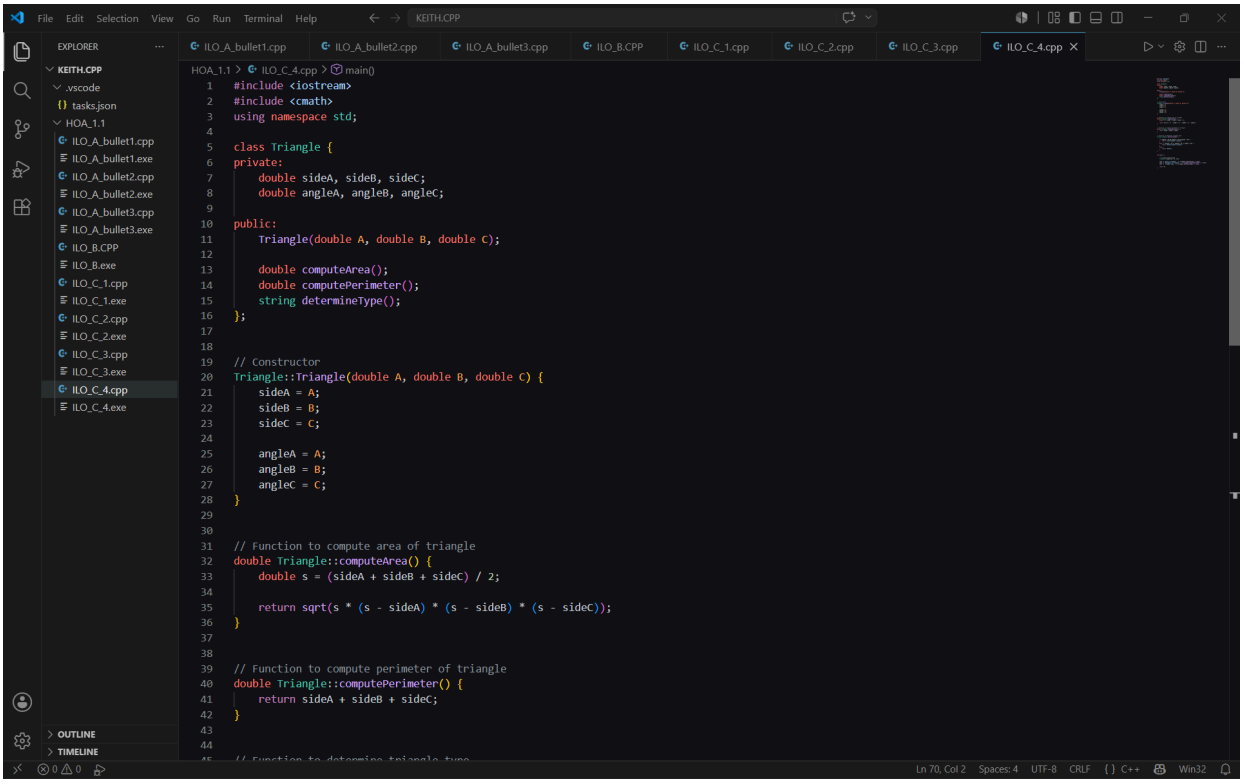
// Function Definition
double calculateDistance(double x1, double y1, double x2, double y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}
```

```
Distance between points: 2.82843
PS C:\Us^Cs\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "c:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1\ILO_C_3.exe"
Enter x1: x1
Enter y1: Enter x2: Enter y2: Distance between points: 0
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```

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```cpp
#include <iostream>
#include <cmath>
using namespace std;

class Triangle {
private:
    double sideA, sideB, sideC;
    double angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);

    double computeArea();
    double computePerimeter();
    string determineType();
};

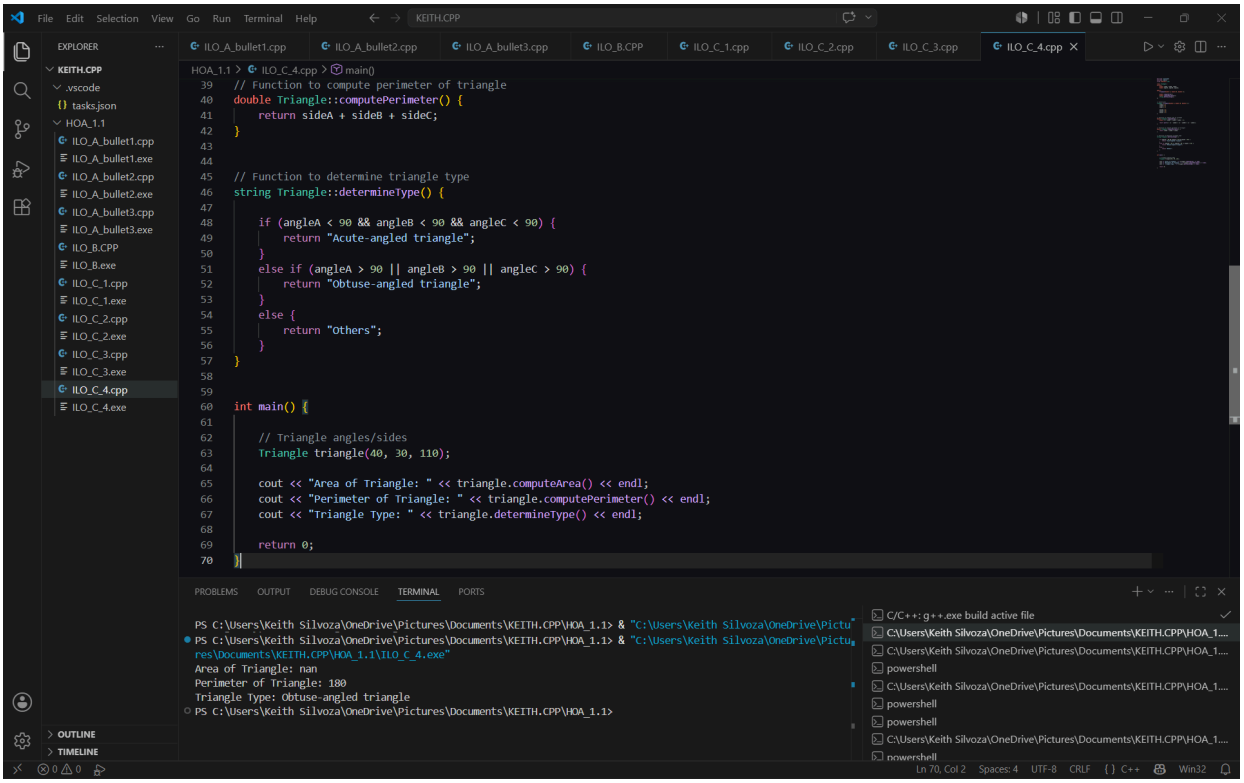

// Constructor
Triangle::Triangle(double A, double B, double C) {
    sideA = A;
    sideB = B;
    sideC = C;

    angleA = A;
    angleB = B;
    angleC = C;
}


// Function to compute area of triangle
double Triangle::computeArea() {
    double s = (sideA + sideB + sideC) / 2;

    return sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}


// Function to compute perimeter of triangle
double Triangle::computePerimeter() {
    return sideA + sideB + sideC;
}
```



```cpp
// Function to compute perimeter of triangle
double Triangle::computePerimeter() {
    return sideA + sideB + sideC;
}


// Function to determine triangle type
string Triangle::determineType() {

    if (angleA < 90 && angleB < 90 && angleC < 90) {
        return "Acute-angled triangle";
    }
    else if (angleA > 90 || angleB > 90 || angleC > 90) {
        return "Obtuse-angled triangle";
    }
    else {
        return "Others";
    }
}


int main() {

    // Triangle angles/sides
    Triangle triangle(40, 30, 110);

    cout << "Area of Triangle: " << triangle.computeArea() << endl;
    cout << "Perimeter of Triangle: " << triangle.computePerimeter() << endl;
    cout << "Triangle Type: " << triangle.determineType() << endl;

    return 0;
}
```

```
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictu"
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1> & "C:\Users\Keith Silvoza\OneDrive\Pictu
res\Documents\KEITH.CPP\HOA_1.1\ILO_C_4.exe"
Area of Triangle: nan
Perimeter of Triangle: 180
Triangle Type: Obtuse-angled triangle
PS C:\Users\Keith Silvoza\OneDrive\Pictures\Documents\KEITH.CPP\HOA_1.1>
```

SECTIONS	ANSWER
Header File Declaration Section	#include <iostream>
Global Declaration Section	using namespace std;
Class Declaration and Method Definition Sectio	class Triangle with private variables totalAngle, angleA, angleB, angelC and public methods
Main Function	Triangle::Triangle(), Triangle::setAngles(), Triangle::validateTriangle()
Method Definition	Creates object Triangle set1 (40,30,110) and calls validateTriangle
Method Definition	Contains the implementation of constructor, setter, and validation method

In the case of problem #4, the values present in ILO B are angles. For area and perimeter, class will prefer using side lengths so depending on what you are expecting your instructor to enter for the values you can change main()

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:
chatgpt, quillbot

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **In this article, i relied on Ai to mainly guide me thru the procedures to create the C++ programs thru cause i had trouble in understanding some aspects. I also used this to remind me once again od codes/syntax.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ai ass a guide to understand the procedures recall C++ codes and fix errors.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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